RECOVERED MANUSCRIPT AND ANALYTICAL REVIEW

The finite-lift existence argument for the complete Q222 quantum set

Recovery, mathematical structure, and review scope

22 October 2026

Result of this recovery

The analytical existence-proof manuscript has been recovered as an actual document with its mathematical sources. The principal recovered release is the **798-page R21 rederivation dated 22 October 2026**, with 215 indexed mathematical components. Its opening states a universal tensor-sum-of-squares theorem and a finite semidefinite-lift conclusion. It explicitly distinguishes itself from the unrecovered original 549-page R21 manuscript.

The August 30 manuscript, all six August 31 corrections, and the later computer-free proof map, first-principles guide, and consolidated insights have also been recovered. Their scopes differ: the August route claims an order-two/SOC representation; the October route claims qualitative finite SDP representability without a block-size bound. The undelivered evaluated pencil in E1–E3 is separate from both theorem-level questions.

Analytical finding. The physical reduction, finite source reduction, two closed-face lifts, positive-residue mechanism, legal normalization rules, and final finite mixture/criterion implication are supported by the written arguments examined here. The later full R1 construction is present. Targeted examination supports its effective normal reserve, unrestricted state ratio and outer testing regimes. The selected C1 endpoint and its local/remote state-dependent join also have substantive written proofs. No specific contradiction was found in the examined October parts.

Review boundary. This is source recovery and a substantial scoped analytical audit, not a fresh independent verification of every identity and every local branch in 798 pages. The unconditional October theorem is the recovered manuscript's conclusion. This report verifies its structure and selected critical arguments while identifying dependencies not rederived in full. No solver, sampled test, symbolic-check receipt, hash, or historical completion label is used as a mathematical premise.

Specific issues were identified in the stronger August proof, including a mixed-denominator assertion and an exact-versus-residue mismatch in a reciprocal coefficient. They are documented below. They neither show that $\mathcal{Q}_{222}$ has no finite lift nor automatically invalidate the separate October construction.

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1 Recovered objects and reading route

Object	Extent	Role
R21 fresh rederivation	798 pages; 215 components	Universal ordinary tensor SOS and qualitative finite SDP-lift existence are the stated conclusions. Full source archive retained.
August base and overlay	2,415-line base; six corrections	Composite frozen source claiming $\mathrm{sxdeg}(\mathcal{Q}_{222}) = 2$, a distinct rank-controlled argument.
Computer-free proof map	21 pages	Dependency map, source geometry and R1 reserve simplification; not a re-audit of all imported bodies.
First-principles guide	13 pages	Foundations, source-only repair of August (9.11), and application obligations.
Consolidated insights	11 pages	Nine-square scalar Laurent certificate, even-power tangent gaps and reserve organization, with stated scope limits.
Current report and audits	Separate new files	Fresh recovery, analytical structure, selected local checks and version-specific findings.

The valid R21 archive contains 4,690 members. All 4,689 files bound by its manifest and all 445 source/presentation/review bindings were checked: zero missing files and zero hash mismatches. The PDF page count was read from the actual file. These establish identity relative to the delivered archive, not mathematical validity.

The original 549-page release remains unrecovered. One older recovery ZIP has an invalid central-directory structure; its received bytes and digest were recorded separately. The intact later rederivation contains its own retained historical sources. It is not a byte-identical recovery of the old release.

The following are *physical PDF page numbers* in the recovered 798-page manuscript, checked against the actual outline.

Page	Content
11	Opening guide and complete intended theorem chain.
14	Universal tensor theorem and finite-lift implication.
16	Positive-residue and strict bounded-Gram mechanism.
23	Canonical physical parent and geometry.
49	Finite induction for noncommuting residue tests.
57	Exact lifts of the two closed source-face types.
61	Finite-contact reduction to C1 and C2.
66	Native coordinates and source-sign entry.
667	Doubly infinitesimal C1 endpoint with bounded β/q^2.
697	Complete canonical local-certificate/source dispatch.
705	Positive-total-scale source union.
766	Later full R1 theorem without a source-ratio split.
794	Directed weak-C2 source and testing join.

The delivered component index preserves all 215 source/presentation bindings and the outline. The original source ZIP contains the complete mathematical bodies, not merely this selected route.

2 The exact existence statement

The set is the full finite-dimensional bipartite tensor-product quantum set in the coordinates

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}) \in \mathbb{R}^8.$$

Writing outcomes as $a, b \in \{-1, 1\}$, its probabilities are

$$p(a, b \mid x, y) = \frac{1 + a\,a_x + b\,b_y + ab\,E_{xy}}{4}.$$

Let P be the cone of affine functions nonnegative on $\mathcal{Q}_{222}$. For a real closed extension $K \supseteq \mathbb{R}$, a subscript K means base extension of the fixed semialgebraic description.

For arbitrary separately labelled real closed extensions K_c, K_t, put

$$\mathcal{T} = K_c \otimes_{\mathbb{R}} K_t, \qquad [f, q] = f_0 \otimes 1 + \sum_{\nu=1}^{8} f_\nu \otimes q_\nu.$$

The source coefficients belong to K_c; the independent testing point belongs to K_t. These are not Alice/Bob labels. The factors may be two labelled copies of the same field. No ordering, field property, or algebraic-disjointness property is assumed for $\mathcal{T}$.

The October manuscript's universal claim is

$$\boxed{[f, q] \in \Sigma(K_c \otimes_{\mathbb{R}} K_t)^2 \quad \text{for every } K_c, K_t,\ f \in P_{K_c},\ q \in (\mathcal{Q}_{222})_{K_t}.} \tag{1}$$

Every certificate is finite. A common number of roots and a rank-two bound are not part of this qualitative statement.

Scheiderer's Corollary 3.19 characterizes closed convex semialgebraic spectrahedral shadows by this tensor-SOS condition for two copies of every real closed extension. Its ambient dimension is arbitrary; the planar main theorem in the paper's title has a different scope. Thus (1), with the finite physical model below, gives finite N, m and real symmetric matrices satisfying

$$\boxed{\mathcal{Q}_{222} = \left\{ q : \exists y \in \mathbb{R}^m, \quad A_0 + \sum_{\nu=1}^{8} q_\nu A_\nu + \sum_{j=1}^{m} y_j B_j \succeq 0 \right\}.} \tag{2}$$

The same paper's Theorem 3.10 adds a uniform bound on tensor-root rank and controls semidefinite block size. That is the stronger criterion used by the August route. [S1]

The external implication is correct. The Q222-specific burden is the universal hypothesis, not evaluating the eventual matrices. Conversely, ordinary scalar nonnegativity alone does not supply that hypothesis.

3 The finite physical model

Put $\mathcal{Z} = S^1 \times S^1 \times S^3$. For $z = ((c_A, s_A), (c_B, s_B), \psi)$, let $\Phi(z)$ be the eight expectations in a real unit state $\psi \in \mathbb{R}^4$, using

$$A_0 = B_0 = Z, \quad A_1 = c_A Z + s_A X, \quad B_1 = c_B Z + s_B X,$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Proposition 3.1 (Exact real-qubit reduction)**.**

$$\mathcal{Q}_{222} = \operatorname{conv}\Phi(\mathcal{Z}) = \left\{ \sum_{i=1}^{9} w_i \Phi(z_i) : z_i \in \mathcal{Z},\ w_i \geq 0,\ \sum_i w_i = 1 \right\}.$$

Consequently $\mathcal{Q}_{222}$ is compact, convex and semialgebraic.

Proof. A binary observable is a Hermitian contraction A, with involutive dilation

$$\widehat{A} = \begin{pmatrix} A & \sqrt{I - A^2} \\ \sqrt{I - A^2} & -A \end{pmatrix}.$$

The same embedding $h \mapsto (h, 0)$ works for both local settings. Applying both local embeddings preserves joint expectations.

For two involutions U, V, $C = UV$ is unitary and $UCU = C^*$. Pairing conjugate eigenspaces gives invariant blocks of dimension two; real eigenvalues give one-dimensional blocks. Local basis changes put each nontrivial pair into the real form above. One-dimensional strategies are represented on common eigenvectors with commuting measurement choices.

Block-diagonal measurements retain only the diagonal state blocks, whose traces are convex weights. A complex pure vector $x + iy$ against a real symmetric observable M satisfies

$$(x + iy)^* M (x + iy) = x^T M x + y^T M y.$$

Separate normalization gives real pure-state atoms. Mixed states are included by pure-state decomposition. Conversely, finite mixtures are realized by local direct sums and a state on matching block pairs.

Carathéodory elimination in $\mathbb{R}^8$ leaves at most nine atoms. The displayed parameter space with simplex weights is compact and semialgebraic, and the expectation map is polynomial. Its image is compact and semialgebraic; direct sums establish convexity. □

This covers complex and mixed states and arbitrary finite original dimensions. The qubit reduction agrees with Masanes's extremal realization theorem. [RFA; S3] The fixed nine-atom formula defines $(\mathcal{Q}_{222})_K$, and

$$f \in P_K \quad \Longleftrightarrow \quad \forall q \in (\mathcal{Q}_{222})_K,\ f(q) \geq 0$$

is a first-order ordered-field formula. Transfer does not assert order-topological compactness of a non-Archimedean field.

4 The algebra that makes the proof computer-free

4.1 Why the two tensor factors must remain separate

In a real closure K of $\mathbb{R}(t)$, ordered with $t > 0$ infinitesimal, the tensor $\Delta = t \otimes 1 - 1 \otimes t$ is nonzero, although multiplication sends it to zero. Extend d/dt to the algebraic extension and apply $a \otimes b \mapsto a\, db/dt$; the image is -1.

A mixed tensor with positive residue need not be a unit. For example $1 + \Delta^2$ is not invertible in $K \otimes_{\mathbb{R}} K$. Embed the second copy of $\mathbb{R}(t)$ into $K(i)$ by $t \mapsto t + i$, and extend to an embedding of its algebraic real closure. The induced homomorphism sends Δ to $-i$, so it sends $1 + \Delta^2$ to zero. A unit cannot have zero image in a field.

Thus a source-field Schur complement can prove source geometry without automatically authorizing the same inverse in the tensor ring. The proof needs pure-factor operations or exact finite tensor identities. [TF]

4.2 Legal restoration, matrix pairing and mixture identities

For $s > 0$ in K_c and $t > 0$ in K_t,

$$(s \otimes t)F \in \Sigma\mathcal{T}^2 \quad \Longleftrightarrow \quad F \in \Sigma\mathcal{T}^2,$$

because the multiplier is the invertible square $(\sqrt{s} \otimes \sqrt{t})^2$.

If $X = BB^T$ over K_c, $Y = CC^T$ over K_t, then

$$\sum_{i,j} X_{ij} \otimes Y_{ij} = \sum_{a,b} \left(\sum_i B_{ia} \otimes C_{ib} \right)^2 . \tag{3}$$

Singular PSD matrices are allowed.

For nonnegative source weights a_j and testing weights w_i with $\sum_i w_i = 1$, write $f = \sum_j a_j f_j$, $q = \sum_i w_i q_i$. If $[f_j, q_i] = \sum_k h_{ijk}^2$, then

$$[f, q] = \sum_{i,j,k} \left((\sqrt{a_j} \otimes \sqrt{w_i}) h_{ijk} \right)^2 . \tag{4}$$

This is finite linearity in f and affineness in q. The testing-weight sum preserves the affine constant. Zero weights are never inverted. [O, FA10–FA12]

4.3 Positive residue gives an exact tensor SOS

Let $\mathcal{O}_c, \mathcal{O}_t$ be the rings of elements bounded by real constants and set $\mathcal{B} = \mathcal{O}_c \otimes_{\mathbb{R}} \mathcal{O}_t$. Their standard-part maps induce a real residue map on $\mathcal{B}$.

Lemma 4.1 (Positive residue)**.** *Every $a \in \mathcal{B}$ with $\operatorname{res}(a) > 0$ is a finite sum of positive pure tensors, hence an SOS with rank-one roots.*

Proof. Expand bounded factors around their real residues:

$$a = \epsilon + \sum_{j=1}^{N} x_j \otimes y_j, \qquad \epsilon > 0,$$

where each term has an infinitesimal factor. Split $\epsilon = \sum_j \epsilon_j$ into positive real reserves. If x_j is infinitesimal, choose real $M_j > |y_j|$. Exactly,

$$\epsilon_j + x_j \otimes y_j = \frac{1}{2} \left[(\epsilon_j / M_j + x_j) \otimes (M_j + y_j) + (\epsilon_j / M_j - x_j) \otimes (M_j - y_j) \right].$$

All four factors are positive in their own fields, so take their separate square roots. Exchange factors when the infinitesimal is y_j. The case $N = 0$ is immediate. □

Lemma 4.2 (Strict bounded Gram)**.** *If $H \in \operatorname{Sym}_r(\mathcal{B})$ has positive-definite real residue, then $v^T H v \in \Sigma\mathcal{T}^2$ for every $v \in \mathcal{T}^r$. The rows v_i need not be bounded.*

Proof. A real congruence makes the residue I. Choose real $\delta > 0$ with $(r-1)\delta < 1$. The exact identity is

$$\begin{aligned} H = & \sum_i [H_{ii} - (r-1)\delta] e_i e_i^T \\ & + \sum_{i<j} \frac{\delta + H_{ij}}{2} (e_i + e_j)(e_i + e_j)^T \\ & + \sum_{i<j} \frac{\delta - H_{ij}}{2} (e_i - e_j)(e_i - e_j)^T . \end{aligned}$$

Every scalar coefficient has positive real residue. Apply the preceding lemma and evaluate on v. □

There is no infinite expansion or mixed square root. The hard application step is producing an exact full-parent Gram with bounded coefficients and *strict* real residue. Positive infinitesimal eigenvalues do not meet the latter condition. [G]

4.4 Regular contact and the additional rank-two condition

If a real polynomial $F(s,t)$ satisfies $F(s,s)=0$ and $\partial_t F(s,s)=0$, then polynomial expansion in $h=t-s$ gives exactly

$$F(s,t)=(t-s)^T H(s,t)(t-s), \qquad H(p,p)=\tfrac{1}{2}\partial_t^2 F(p,p).$$

No higher term is discarded. When independent source and test points specialize to p and $H(p,p)\succ 0$, strict Gram applies.

For a smooth affine variety, the analogous statement uses an exact tensor in the square of the diagonal ideal and a positive intrinsic quadratic class. Actual separate lifts to a common valid chart are needed. Scheiderer's diagonal lemma supplies this mechanism under its precise hypotheses. [TF; S2]

Every real linear combination of genuine divided differences $f_i(s)\otimes 1-1\otimes f_i(t)$ is again one divided difference, so has rank at most two. Multiplication by rank-one coefficient roots preserves this bound. Arbitrary rows that individually have rank two need not have a real span consisting of rank-two tensors. The October qualitative route requires finite root rank, not the additional uniform bound two.

5 The exact source funnel

For a fixed CHSH orientation,

$$H=1-\tfrac{1}{2}(E_{00}-E_{01}+E_{10}+E_{11}), \qquad f=dH+4\sum_{\xi}\xi p_\xi,$$

with eight nonnegative event coefficients $\xi=\lambda,n,m,C,A,D,\alpha,\beta$. For $D_u=1+u^2$, $D_v=1+v^2$, and state (W,X,Y,Z), the complete parent is

$$\begin{aligned}F=\;&4\lambda D_u(W+vX)^2+4nD_uD_vX^2+4m(vW-X+uvY-uZ)^2\\&+4CD_v(uW-Y)^2+4AD_uD_vY^2\\&+(4D+2d)(uW+uvX-Y-vZ)^2\\&+(4\alpha+2d)D_u(vY-Z)^2+(4\beta+2d)D_v(X+uZ)^2-2dD_uD_vZ^2.\end{aligned}\tag{5}$$

Coefficient and testing expressions occupy separate factors. A scalar restriction of (5) cannot replace a certificate for the entire state form.

For homogeneous measurement pairs, with $\Delta_A=a_0^2+a_1^2>0$, $\Delta_B=b_0^2+b_1^2>0$,

$$\mathcal{F}=(1\otimes\Delta_A\Delta_B\|\psi\|^2)[f,\Phi(z)].$$

The factor is pure testing and positive for a nonzero state. The homogeneous identity includes literal endpoints without dividing by a vanished affine coordinate. [O, FA4–FA6; RFA]

5.1 Reduction to events, closed faces and two contact families

A valid affine functional either is nonnegative on the nonsignaling polytope and is a nonnegative event sum, or is negative at a PR vertex. That PR vertex and its eight neighboring deterministic vertices form an affine evaluation basis, giving the canonical coefficient representation above.

For $d > 0$, the signed $Z = 0$ restriction is

$$2d\{(uW + uvX - Y)^2 + D_u v^2 Y^2 + D_v X^2\} \succ 0 \quad (uv \neq 0).$$

Thus the three-state principal block is positive definite. The full state determinant is

$$\det P = 16 D_u^2 D_v^2 u^2 v^2 L(U, V), \qquad U = u^2,\ V = v^2,$$
$$L(U, V) = h + \sum_{e \in \{-1,0,1\}^2 \setminus \{0\}} q_e U^{e_1} V^{e_2}, \qquad q_e \geq 0.$$

The source body prints all eight nonnegative factorizations. At a positive stationary contact, $a_e = q_e U_c^{e_1} V_c^{e_2}$ obey

$$h = -\sum_e a_e, \qquad \sum_e a_e e = 0, \qquad \mathcal{H}_L = \sum_e a_e e e^T.$$

Rank-zero and rank-one support possibilities are excluded by explicit event-incidence arguments and negative physical tests. This gives coercivity and strict *single-field* curvature at every actual finite zero.

Conversely, nonnegative coefficients, a finite kernel state and both angle jets zero imply physicality: the convex logarithmic Laurent profile has a global minimum there. Logarithms are used over $\mathbb{R}$; the resulting finite ordered-field implication transfers. None of this inverts a tensor or promotes a source-field Hessian to strict residue. [SE1–SE10]

Normalize the coefficient sum to one over $\mathbb{R}$. A coercive extreme source cannot have a strictly positive Laurent minimum, since small opposite coefficient perturbations preserve positivity. Thus it has a finite contact. There are four kernel equations and two angle equations, linear in the nine coefficients. Seven or eight positive events would permit a nonradial sign-preserving perturbation retaining all six equations, contradicting extremality. Hence five or six events are positive.

The compatible omission patterns reduce by actual Bell symmetries to

$$\begin{aligned} &\text{C1}: \quad d, m, C, A, \alpha > 0, \quad \lambda = n = 0, \quad D, \beta \geq 0, \quad D + \beta > 0, \\ &\text{C2}: \quad d, m, C, A, D, \alpha > 0, \quad \lambda = \beta = 0, \quad n \geq 0. \end{aligned}$$

Incompatible omissions enter two closed-face types. The compact normalized real source base has affine dimension at most eight, so nine extreme sources suffice. This fixed finite assertion then transfers to every source field. [SE11–SE15]

5.2 The closed faces have explicit lifts

For a symmetric quadratic matrix polynomial $K(t) = K_0 + tK_1 + t^2 K_2$, the retained argument proves

$$K(t) \succeq 0 \ \forall t \iff \exists S = -S^T : \quad \begin{pmatrix} K_0 & K_1/2 + S \\ K_1/2 - S & K_2 \end{pmatrix} \succeq 0.$$

The proof uses a closed Gram cone, a finite matrix-moment decomposition and separation, including singular cases. It yields one 6×6 block for $\lambda = m = 0$, and two 3×3 blocks for $\lambda = \alpha = 0$.

Both have displayed strictly feasible points. Strict feasibility bounds the trace of PSD witnesses when their adjoint images stay bounded, so the adjoint image is closed and the evaluation witness is attained. The finite witness statement transfers to the testing field, and (3) gives tensor squares. This is an actual analytical closed-face input. [CF1–CF11; RFA]

6 Finite closure and existence

The remaining local proposition must certify the complete parent for every actual physical C1/C2 source, every independent projective test, every state, all allowed literal zero coefficients and all positive openings. The initial scale

$$\delta_{\mathrm{tot}} = \frac{d}{d+\lambda+n+m+C+A+D+\alpha+\beta}$$

is split before a $d = 1$ gauge. Positive-unit and positive-infinitesimal total scale are distinct cases; $d = 1$ does not itself imply bounded event coefficients.

Assuming this full local proposition, each event source has a direct projector certificate, each closed-face source uses matrix pairing, and each C1/C2 source uses its local parent identity. If $(s \otimes 1)\mathcal{F} = \sum_k h_k^2$, $s > 0$, restoration gives

$$[f, \Phi(z)] = \sum_k \left[\left(s^{-1/2} \otimes (\Delta_A \Delta_B \|\psi\|^2)^{-1/2} \right) h_k \right]^2 .$$

All inverses are pure field operations. The finite source decomposition and nine-atom testing decomposition, combined with (4), give (1). Taking two copies of one arbitrary real closed field then yields (2). [O; LC9–LC10]

This final implication is sound with those hypotheses. E1–E3 ask for an evaluated common feature list, a concrete affine pencil and projection, and both-inclusion validation of its data. They are downstream constructive outputs, not extra hypotheses of the qualitative criterion.

7 Targeted analysis of the recovered local proofs

The current review includes four complementary October audits: global/source assembly and a delicate C1 endpoint; the later full R1 proof; noncommuting-residue constrained geometry; and C2 source dispatch. The separate audit notes give exact paths, equation tags, hypotheses and limits. Their findings are summarized here without converting scoped checks into certification of all 215 components.

7.1 R1: the effective reserve and the whole parent

The later R1 theorem expressly removes the earlier restriction that jT be infinitesimal. Its compact source variables satisfy

$$0 < p < \min(1, j), \quad 0 < H < 1, \qquad B, T, jB, TB \text{ positive infinitesimals},$$

while j and jT need not be bounded. A literal inverse map to the original stationary source is printed.

For $h_0 > 0$, $n \geq 0$, the central exact completion is

$$h_0 N_0^2 + n(N_0 + jb)^2 = (n + h_0) \left(N_0 + \frac{nj}{n+h_0} b \right)^2 + \frac{n h_0 j^2}{n+h_0} b^2. \tag{6}$$

The denominator is strictly positive and pure source. Writing

$$c_b = \frac{nj}{n+h_0}, \qquad M_e = \frac{n h_0 j^2}{n+h_0},$$

the usable reserve is M_e, not the uncompleted mass nj^2. At $n = 0$, both quantities vanish literally. For $n > 0$,

$$\frac{c_b^2}{M_e} = \frac{n}{h_0(n + h_0)} \le \frac{1}{h_0},$$

which controls the shift against the retained current.

The full-normal proof retains both unrestricted state normals, both first-normal arrays, and their quadratic cross term. Scalar, radial and circuit identities buy helper pivots; strict Gram then applies to the remaining complete expression. The scalar restriction alone would not suffice. [R1N; FN]

The contact rows use

$$U = \frac{u}{\sqrt{jB}}, \quad V = \frac{v}{\sqrt{TB}}, \quad Z_* = \frac{z}{\sqrt{jT}},$$

and the exact circuit

$$\begin{aligned}\Psi &= U^2 + V^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVZ_* \\ &= (U - V)^2 + Z_*^2(1 - UV)^2 + 2UV(1 - Z_*)^2.\end{aligned}$$

The rows can be unbounded. The positive state ratio z remains arbitrary inside them. The square root of UV factors into pure-source and pure-testing quantities.

The opposite second-normal contribution is treated through the whole squared multiplier

$$T(jT)v^2(\alpha - BTCHu^2)^2.$$

The newer cost estimates cancel jT exactly and retain all higher terms and the cross term. They do not compare source and testing infinitesimals. [FN24; R1N11]

For outer tests put

$$C_t = v(uX - Z), \qquad B_t = X + uZ, \qquad g_* = \frac{CD}{C + D}.$$

A pure-source harmonic completion gives $g_*C_t^2$. The exact relation

$$uB_t - C_t/v = (1 + u^2)Z$$

produces the two-row outer Gram when v is noninfinitesimal. The small-v four-row identity and the remaining six-row identity retain every state coordinate. Real cap choices use positivity on event kernels; excess coefficients stay in whole event squares.

The audit directly checked these outer identities, including the zero row relation used in the six-row adjustment. The state/sign join covers $wz \le 0$ with a pure-testing nonnegative core, and normalizes $w = 1$ when $wz > 0$ while keeping $z > 0$ arbitrary. Homogeneous formulas include literal endpoints, and complex states split into real and imaginary parts. [R1N12–R1N24; SJ]

Finding. The later R1 argument is present and substantive. No concrete gap was found in the examined allocation and joins. Older records marking R1 open should not be substituted for this later body. The large source-polynomial, scalar-circuit and radial coefficient expansions were not all independently recomputed in this review.

7.2 A shorter R1 reserve formulation

The later proof map combines several source cases using

$$\rho = Bj^2, \quad \delta = 1 - p, \quad S = \delta + \rho + M_e, \qquad p^2 \le 10M_e + 4\rho, \quad \frac{1}{10} \le S < 9.$$

Weighting the exact budgets by $\delta/S, \rho/S, M_e/S$ avoids isolated inverse openings. S is a positive pure-source unit, and $M_e = 0$ is retained without division by M_e. This is a useful shorter exposition, retaining the full parent, circuit, radial and normal identities as inputs. [RM]

7.3 The C1 JC4/SN endpoint

The reviewed endpoint has S, κ, ν infinitesimal, $\ell = 1 - \nu$, $q = \nu/\ell$, and $n_s = \beta/q^2$ bounded. The exact source identities give

$$b_0^2 = O(\kappa\theta n_s), \qquad \frac{D}{Sq^2} \longrightarrow \frac{1}{2}, \qquad \frac{e}{\kappa} \longrightarrow 1, \qquad \frac{h}{\nu} \longrightarrow 1.$$

This retains literal $\beta = 0$ and excludes $D = 0$. No ordering among the independent infinitesimal scales is assumed. The local $e = xh/\sigma = \kappa h/\nu$ is explicitly defined and is not the earlier native parameter bearing the same letter.

The complete local parent keeps unrestricted state normals X_e, N_y. Its exact relation

$$R = \frac{\kappa\sqrt{S}\,G - uN}{\ell}$$

puts every R-term into an infinitesimal Gram perturbation. The helper identity has the usable form

$$\text{Cost} = \ell^2(wB_u)^2 + (\kappa\delta W)^2,$$

where the cost's coefficients on strict base rows have zero residue. It buys any fixed positive helper pivot on wB_u, together with a genuine square, at infinitesimal cost. The analogous wB pivot handles the remaining bounded cross terms. When $n_s = 0$, its zero row is omitted.

The remote coverage has three genuine cases. Noninfinitesimal Alice tests use complete four-state Grams since C, A are unbounded. Infinitesimal Alice with unmatched Bob uses a homogeneous six-root Gram with scalar-block determinant

$$\frac{1}{4}(\bar{s}^2 - \bar{\ell})^2.$$

On the SN source $\bar{\ell} = 1$; the remaining matched Bob tests have c infinitesimal and s a unit, so $w = c/s$ is infinitesimal, exactly the local SN domain. State ratios remain in roots throughout.

Finding. The source transition, full-state allocation and local/remote union are coherent on the examined formulas. Some coefficient expansions and the other C1 endpoint cells were not independently rederived. [Current assembly audit]

7.4 Noncommuting residue tests

The A6 route uses the geometry of zero events at an entangled residue: the supported zero events form a matching in the event eight-cycle. At most three are independent state-normal coordinates. A fourth matching event enters a separate terminal.

At each source scale, impose the current hard event amplitudes *exactly* zero in the coefficient field, divide by the positive pure-source next scale σ, and then take residue. Physicality makes the resulting constrained residue nonnegative. This does not delete actual terms from the final tensor expression. The weighted-contact lemma returns to the full parent with all normal deviations retained.

If the signed coefficient becomes visible, the one-, two- or three-hard-event constrained curvature argument applies. Otherwise another matching edge is added. The process terminates because there are at most four matching edges, not because an arbitrary infinitesimal series is assumed to stabilize.

The focused audit checked all eight one-hard derivative coefficients and all 24 two-hard activation conditions from the printed determinant dictionary, the explicit negative witness states and the nonzero signed-core differential on the residual matching. The weighted and unequal-scale lemmas

use only real residue Jacobian inversion and source-field implicit-function transfer; they do not invert a mixed tensor.

Finding. No defect was found in the reviewed geometry, constrained curvature and induction interface. The original full determinant identity and the complete arbitrary-scale four-matching terminal remain named mathematical inputs. A6 does not cover commuting-angle residues. [Current A6 audit]

7.5 C2 source dispatch and boundary logic

The HJ source partition reaches ten rows R1–R10 or four regular exits. The two excluded intersections have source-equation justifications:

$$\text{all six weights bounded} \iff t, h, \ell, j, r, g \text{ are units},$$

contradicting the weak total scale, and the weak-C criterion excludes exactly the simultaneous-small r, j intersection.

The formal C1 comparison has

$$\beta_1 = -\frac{r\delta n}{eP_n}.$$

At $n = 0$ it is an actual C1 boundary source and the precise local C1 theorem is used. At $n > 0$, the comparison has negative β_1; the proof instead checks finite coefficient identities. It does not invoke physical C1 positivity for this formal source.

The complete CT reflection preserves the strict native gaps. For example,

$$g' = \frac{a^2 j}{b}, \qquad j' = \frac{g}{b},$$

$$1 - r' = \frac{b - a^2 zr}{b}, \qquad 1 - z'h' = \frac{a^2 t}{b}, \qquad b' - a^2 z'r' = \frac{a^2(1-r)}{b}.$$

All are positive by the original source domain. Both $n = 0$ and the total-scale predicate are preserved.

The nine ordered angle types and complementary-event cases are exhaustive. The single-angle argument explicitly includes additional opening extensions beyond its earliest theorem header. The inspected dispatch does not call the entire C1 theorem in a circle; its C1 imports are named local bodies.

Finding. No uncovered source cell, false reflection, lost $n = 0$ boundary or whole-theorem routing cycle was found. Native entry, positive-scale coverage, A6 and the detailed local conductors remain substantive inputs; dispatch verification is not independent rederivation of every leaf. [Current C2 dispatch audit]

8 Specific findings in the August rank-two proof

The August base and all six replacements must be read together. The general diagonal mechanism is sound under its hypotheses. The issue is showing that every required quantum contact has the actual lifts, exact identities, strict real residues and rank control needed for its application. The full August audit is delivered separately.

8.1 The mixed denominator in (9.11)

The printed completion uses

$$e(1+u^2)+\beta$$

and says the denominator lies wholly in one tensor factor. Under the manuscript's own typing, e, β are source parameters and u is a testing parameter. That assertion is false as written.

For its pair $e(1+u^2)X^2+\beta(X+uB)^2$, a source-only repair is, with $c=e+\beta>0$,

$$c\left(X+\frac{\beta}{c}uB\right)^2+\frac{e\beta}{c}u^2B^2+eu^2X^2. \tag{7}$$

Expansion proves equality. If $c=0$, nonnegativity gives $e=\beta=0$, and the pair is zero. For testing X, u, B, the roots have ranks at most $2,1,1$. The last term must remain: with $N=X+(\beta/c)uB$, it becomes

$$eu^2N^2-2e(\beta/c)u^3BN+e(\beta/c)^2u^4B^2.$$

A pairwise repair does not by itself repair the subsequent exact critical center and all degenerate allocations. [A0, §9.1; FP]

8.2 A reciprocal exact-versus-residue mismatch

Equations (13.0o)–(13.0q) give

$$\frac{d}{\lambda}=2\zeta U_d, \qquad U_d=\frac{JN_d}{D_0D_1}, \qquad \overline{U_d}=1.$$

After setting $\lambda=1$, $z=\varepsilon/c=\zeta$, the later equation (13.2) calls $d=2z$ a complete contact equation. A unit with residue one need not equal one.

Appendix A makes the discrepancy checkable by hand. Put $x=r^2$. The constant terms of D_0, D_1, N_d are J, so the coefficient of x is

$$\begin{aligned}[x](JN_d-D_0D_1) &= J(N_1-D_{0,1}-D_{1,1})\\ &= J\left[J^2(1+z)+Kc^2\right]>0\end{aligned}$$

on the real strict chamber. Thus $JN_d=D_0D_1$ is not a polynomial identity. The exact formula has the nonconstant factor U_d; $d=2z$ must be identified as an initial-layer formula or supplied with an exact reparametrization and restoration argument.

This identifies an inconsistency in the printed identities' scope. It is not a quantum counterexample and does not establish failure of the separate October formulas. [A0, §13 and Appendix A; current August audit]

8.3 Exact children and real cotangent validity

Further application obligations remain explicit in the audit:

- Zero residue is not literal zero. A positive infinitesimal coefficient remains a term in the parent. A child stratum requires an exact reduction or a collar that retains transverse terms.
- The lexicographic invariant $(N_{\mathrm{act}}, d_{\mathrm{stratum}})$ proves termination only after every genuine recursive arrow is shown to decrease it. Loss of a positive residue alone decreases neither entry.
- In terminal F_3, $z_2=R^4p^4(1+Wh)/h^2$ has $dz_2=0$ at the real residue $R=0$, while the proposed cotangent change contains $R/(4z_2)$. Invertibility for nonzero valued-field R is not invertibility of the ordinary cotangent map at the real boundary.

- A Laurent layer $A\tau^2 + B\tau^{-2} + C$ with an interior zero has the stated strict curvature only after the identically zero layer is excluded or routed to a retained next layer.

These concerns belong to the August theorem application, not to E1–E3. They do not negate the general diagonal lemma. They explain why the August order-two conclusion should not be imported into the October qualitative route solely from a lock label.

9 Precise status and next verification obligations

ID	Status in this recovery	Exact finding or remaining task
R1	Recovered	Complete 798-page manuscript, original R21 source ZIP and 215-component index. The historical 549-page object remains unrecovered.
R2	File identity checked	ZIP integrity, all 4,689 bound files and 445 dependency bindings checked without mismatch. These are not mathematical certificates.
R3	Recovered	August base, all six corrections and the three later guides; original files preserved.
A1	Analytically checked	Finite physical model, nine-atom convexification, compactness, semialgebraicity and transfer interface.
A2	Analytically checked	Pure-factor restoration, matrix pairing, positive residue, strict bounded Gram and finite mixture closure.
A3	Examined analytically	Finite-contact source reduction, closed faces and native entry; no concrete defect found in this pass.
A4	Scoped positive review	Later R1 full-parent mechanism, outer identities and state/endpoint join. Large coefficient expansions remain named inputs.
A5	Scoped positive review	JC4/SN source fit and full-state local/remote join. Other C1 cells and some expansions were not independently rederived.
A6	Scoped positive reviews	Noncommuting constrained geometry and C2 dispatch. Exact checked steps and inherited premises are listed in the notes.
A7	Conditional implication checked	Complete universal C1/C2 parent certificates imply universal slack SOS and qualitative finite SDP-lift existence.
A8	Not fully repeated	Independent verification of every remaining local identity, normalization, remote test and source boundary in all 215 components. This is an audit workload, not a demonstrated mathematical gap.
B1	Pairwise correction available	August (9.11) has a source-only replacement; propagate its full retained higher term through the subsequent branch.
B2	Clarification/repair required	Restore U_d in August §13 or justify a complete exact reparametrization and restoration.
B3	Application proof required	August common-child lifts, real-residue cotangent validity and strict decrease for all actual descent arrows.

ID	Status in this recovery	Exact finding or remaining task
E1–E3	Separate constructive work	Evaluate the global feature/dimension list, affine pencil/projection, and both-inclusion proof for those actual matrices. These are not premises of qualitative existence.

A complete independent audit should give every leaf its exact source predicate, testing domain, full-parent identity, legal factors and boundary restoration. It should check all invoked identities and prove that the actual source dispatch is exhaustive and acyclic. The existing index and the present focused reviews make that task concrete. They do not replace it with status counts.

The supported recovery conclusion is therefore: **a real analytical existence-proof manuscript has been recovered, its central architecture and selected difficult components have been examined with positive findings, and the precise residual audit scope has been preserved.** The stronger August route has separate, specific repair obligations. No claim that $\mathcal{Q}_{222}$ has no finite lift follows from these findings.

10 Source key and reproducibility

All paths in the machine-readable component index refer to unchanged sources inside the recovered R21 archive. The table below gives short keys used in this report.

Key	Source
O	R21 `publication/opening_assembly.tex`, FA1–FA13.
RFA	`fresh_physical_normalization_conditional_assembly.tex`.
G	`bridge/strict_gram.tex`.
SE	`finite_contact_source_reduction_REDERIVED.tex`.
CF	`closed_faces_short.tex`.
LC	`complete_source_routing_REDERIVED.tex`.
R1N	`r1_all_sources_all_tests.tex`.
FN	`fn_reconstructed.tex`.
SJ	`full_state_sign_join_REDERIVED.tex`.
RM	Computer-free map, `latex/r1_simplification.tex`.
TF	Computer-free map, `latex/tensor_foundations.tex`.
A0	August 30 `Q222_SXDEG2_CONSOLIDATED_PARCHMENT_FINAL`, used with the six August 31 replacements.
FP	First-principles guide and `Q222_9_11_Source_Only_Replacement.tex`.
S1	C. Scheiderer, *Second-order cone representation for convex sets in the plane*, arXiv:2004.04196v2, Theorem 3.10 and Corollary 3.19. https://arxiv.org/pdf/2004.04196
S2	C. Scheiderer, *Smooth hyperbolicity cones are second-order cone representable*, arXiv:2509.17121v2, Proposition 3.11 and Lemma 3.12. https://arxiv.org/pdf/2509.17121
S3	Ll. Masanes, *Extremal quantum correlations for N parties with two dichotomic observables per site*, quant-ph/0512100. https://arxiv.org/abs/quant-ph/0512100

Review method and preservation

Method. The mathematical review used direct reading, analytical derivations and selected hand expansions. No numerical solver or symbolic checker was used as a proof premise. File hashing, archive verification and PDF rendering concern identity and presentation only.

Scope of the reviewers. The focused audits were performed by additional agents within the same model system, alongside the lead review. They are not external peer review or proof-assistant formalization. The individual notes state which mathematical inputs were checked and which were retained.

Preservation. The recovered PDF and source ZIP, August composite source, and three guides remain unchanged. This report and the new audit files are separate. The complete deliverable includes those sources, so the reader can examine the arguments behind every finding.